Q1. Explain Data Wrangling Methods with Suitable Example. (6 Marks)

Introduction to Data Wrangling

Data Wrangling is the process of **cleaning, transforming, integrating, and organizing raw data into a structured and usable format for analysis**. In real-world applications, data collected from different sources is often **incomplete, inconsistent, noisy, or unstructured**. Such raw data cannot be directly used for data analysis or machine learning models.

Therefore, data wrangling is performed to **prepare the data so that meaningful insights can be extracted from it**. It is an important step in the **data preprocessing stage of the data science life cycle**.

For example, a company may collect customer data from **websites, mobile applications, and sales records**. These datasets may contain missing values, duplicate records, or inconsistent formats. Data wrangling helps convert this raw data into a clean and consistent dataset.

Methods of Data Wrangling

Data wrangling involves several important methods such as **data cleaning, data integration, data reduction, data transformation, and data discretization**.

1. Data Cleaning

Data cleaning is the process of **removing errors, inconsistencies, and noise from the dataset**. Raw data often contains missing values, duplicate entries, or incorrect data.

Common techniques used in data cleaning include:

- Handling missing values
- Removing duplicate records
- Correcting incorrect values
- Detecting and removing outliers

Example:
If a dataset contains missing values in the **age column**, they can be replaced with the **mean or median age**.

Data cleaning improves the **quality and reliability of data**.

---

2. Data Integration

Data integration is the process of **combining data from multiple sources into a single unified dataset**.

Organizations often store data in different systems such as databases, spreadsheets, and cloud storage. Data integration merges this data to provide a **complete view of the information**.

Example:
A company may combine:

- Customer data from a website
- Purchase data from a sales system
- Feedback data from surveys

These datasets are integrated to create a **comprehensive customer profile**.

3. Data Reduction

Data reduction is the process of **reducing the size of the dataset while maintaining its important information**. Large datasets require more storage and processing power, so reducing the amount of data helps improve efficiency.

Some common techniques include:

- Dimensionality reduction
- Data compression
- Aggregation

Example:
A dataset containing **100 features** may be reduced to **20 important features** using dimensionality reduction techniques.

---

4. Data Transformation

Data transformation converts data into a **suitable format for analysis or machine learning algorithms**.

This process may include:

- Normalization
- Scaling
- Encoding categorical data
- Converting data types

Example:
Converting temperature values from **Celsius to Fahrenheit** or converting text categories like **"Male/Female" into numerical values (0/1)**.

Data transformation ensures that data is **consistent and compatible with analytical models**.

5. Data Discretization

Data discretization is the process of **converting continuous numerical data into discrete categories or intervals**.

This method simplifies complex data and makes it easier to analyze.

Example:
Age values can be divided into groups such as:

- 0–18 → Child
- 19–40 → Adult
- 41–60 → Middle age
- 60+ → Senior citizen

Discretization is often used in **classification and decision tree algorithms**.

Example of Data Wrangling in Practice

Consider a dataset of **online shopping customers** containing the following problems:

- Missing values in the income column
- Duplicate customer records
- Age stored in different formats
- Large number of irrelevant features

Data wrangling would involve:

1. Cleaning missing and incorrect values
2. Integrating data from different sources
3. Reducing unnecessary features
4. Transforming data into a consistent format
5. Discretizing age into meaningful categories

After performing these steps, the dataset becomes **clean, organized, and ready for analysis**.

After processing, the dataset is checked to ensure that it is **accurate, consistent, and ready for analysis.**

Why Data Wrangling is Needed

Data wrangling is necessary for several reasons:

1. Improves Data Quality

Raw data often contains errors and inconsistencies. Data wrangling improves the **accuracy and reliability** of the data.

2. Enables Better Analysis

Clean and structured data helps analysts and data scientists perform **effective analysis and modeling**.

3. Saves Time in Data Processing

Well-organized data makes it easier to perform queries, visualizations, and machine learning tasks.

4. Improves Machine Learning Performance

Machine learning algorithms require **clean and well-structured data** to produce accurate predictions.

5. Handles Large and Complex Data

Data wrangling helps manage large datasets coming from multiple sources and formats.

Q2. What is Data Discretization? Explain Forms / Techniques of Data Discretization. (6 Marks)

Introduction to Data Discretization

Data Discretization is the process of **converting continuous numerical data into a finite number of discrete categories or intervals**. In many datasets, numerical values may have a very large range and may be difficult to analyze directly. Discretization simplifies such data by grouping values into meaningful intervals.

In data science and data preprocessing, discretization helps **reduce complexity, improve data understanding, and make certain algorithms work more effectively**.

For example, consider a dataset containing the **age of people**:

18, 21, 25, 30, 35, 42, 50, 60

Instead of using exact values, we can convert them into categories such as:

- 0–20 → Young
- 21–40 → Adult
- 41–60 → Middle Age
- 60+ → Senior

This conversion of continuous values into groups is known as **data discretization**.

Discretization is widely used in **data mining, machine learning, and statistical analysis**, especially in algorithms such as **decision trees and classification models**.

Techniques / Forms of Data Discretization

Several techniques are used to perform data discretization. The most common methods include **binning, histogram analysis, clustering, decision tree analysis, and entropy-based discretization**.

---

1. Binning Method

Binning is one of the simplest techniques used for discretization. In this method, continuous values are **sorted and divided into a number of equal-sized groups called bins**.

There are different types of binning:

- **Equal width binning** – each bin has the same interval size
- **Equal frequency binning** – each bin contains the same number of data points

Example:

Marks of students:

35, 40, 45, 50, 55, 60, 65, 70

These values can be grouped into bins such as:

- 30–40
- 41–50
- 51–60
- 61–70

Binning helps smooth noisy data and makes analysis easier.

2. Histogram Method

A histogram is a graphical representation used to show the **distribution of numerical data by dividing it into intervals**.

In histogram-based discretization, data is divided into intervals and represented using bars. Each bar shows the **frequency of values in that interval**.

Example:

Customer age distribution may be grouped as:

Age Group	Number of Customers
0–20	50
21–40	120
41–60	80

Histograms help identify **patterns and frequency distribution** in the data.

3. Clustering Method

In clustering-based discretization, data values are grouped based on **similarity**. Instead of fixed intervals, clustering algorithms automatically find groups of similar values.

Example:

Income values of customers may be grouped into clusters such as:

- Low income
- Medium income
- High income

Algorithms like **K-Means clustering** are often used for this purpose.

This method is useful when the distribution of data is **not uniform**.

---

4. Decision Tree Method

Decision tree algorithms can also be used for discretization. In this method, the dataset is divided into intervals based on **decision rules** that maximize classification accuracy.

Decision trees split data based on **attribute values**, creating intervals that help classify the data more effectively.

Example:

A decision tree may divide age as:

- Age < 25
- Age between 25–45
- Age > 45

These intervals are used for predicting outcomes such as **loan approval or customer purchase behavior**.

5. Entropy-Based Discretization

Entropy-based discretization uses the concept of **information gain from information theory**. It divides data into intervals in such a way that **uncertainty or randomness in the dataset is minimized**.

The algorithm selects split points that **maximize information gain** and improve classification accuracy.

Example:

When building a classification model for customer purchasing behavior, entropy-based discretization can determine the **best intervals of age or income that separate buyers and non-buyers**.

This technique is commonly used in **machine learning algorithms such as decision trees**.

Advantages of Data Discretization

1. **Reduces data complexity**
2. **Improves data understanding and interpretation**
3. **Reduces noise in the dataset**
4. **Improves performance of certain machine learning algorithms**
5. **Makes patterns easier to identify**

Conclusion

Data discretization is an important preprocessing technique that converts continuous numerical values into discrete intervals or categories. Techniques such as **binning, histogram analysis, clustering, decision tree methods, and entropy-based methods** are commonly used to perform discretization. This process simplifies data, reduces complexity, and improves the effectiveness of data analysis and machine learning models.

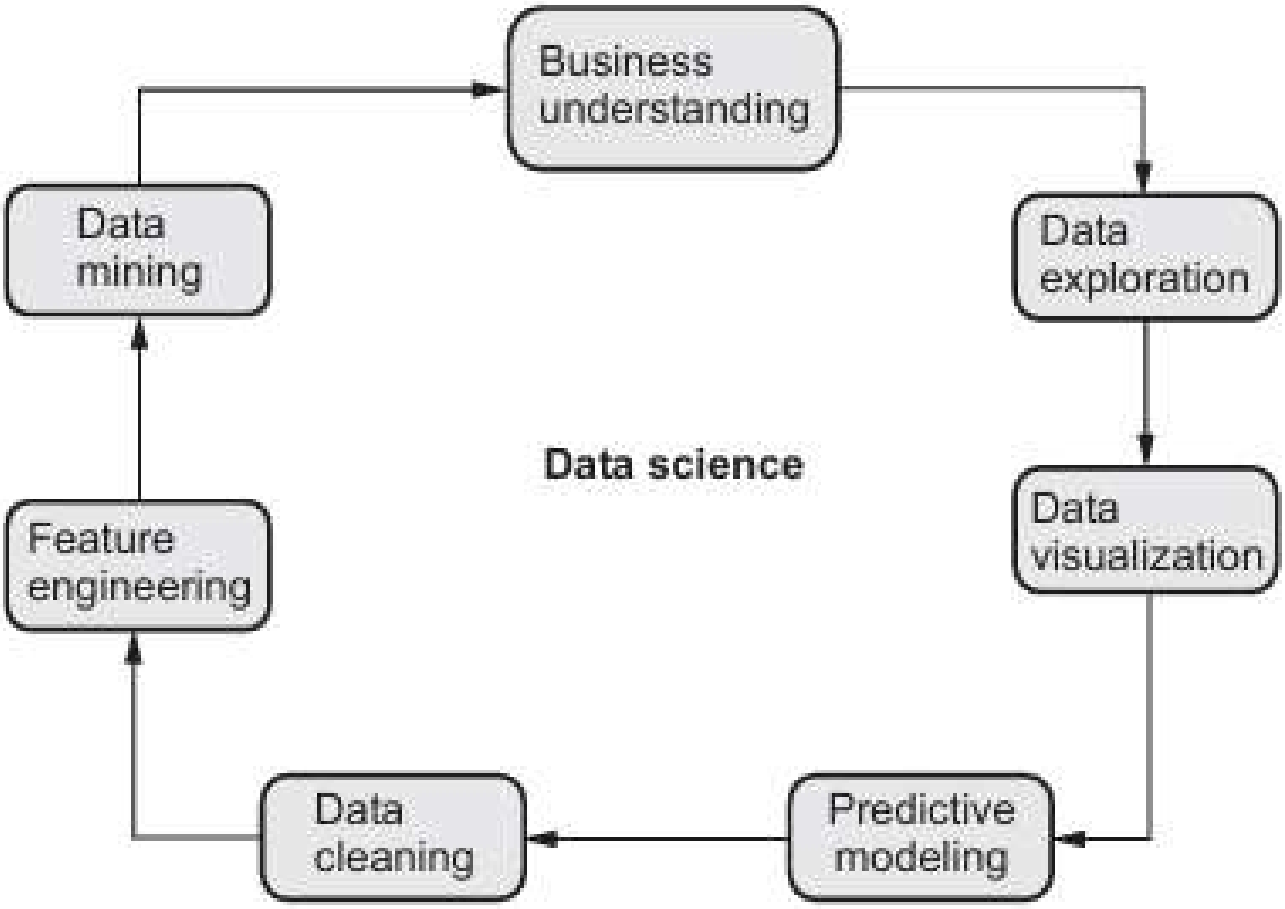

Fig. 1.5.1 Data science life cycle

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Data Science Life Cycle (According to the Diagram)

The **Data Science Life Cycle** is a structured process used by data scientists to analyze data and extract meaningful insights. It consists of **seven important stages** that transform raw data into useful information for decision making.

The process is **cyclical**, meaning after one cycle is completed the process can repeat to improve results.

Phases of Data Science Life Cycle

1. Business Understanding

This is the **first step** where the problem to be solved is clearly defined.

In this stage, the data scientist works with business stakeholders to understand:

- Business goals
- Problem requirements
- Expected outcomes

Example:
An e-commerce company wants to **predict which customers are likely to buy a product**.

2. Data Exploration

In this phase, the collected data is analyzed to understand its **structure, patterns, and relationships**.

Activities include:

- Checking data distribution
- Identifying patterns
- Detecting anomalies or missing values

Example:
Analyzing customer purchase history to find buying patterns.

---

3. Data Visualization

Data visualization helps represent data in **graphical form** so that patterns and trends can be easily understood.

Common visualization tools include:

- Bar charts
- Line graphs
- Pie charts
- Histograms

Example:
A bar chart may show **monthly sales trends** of a company.

---

4. Predictive Modeling

In this stage, machine learning algorithms are used to build **models that can predict future outcomes**.

Examples of algorithms used:

- Regression
- Decision Trees
- Classification algorithms
- Clustering algorithms

Example:
A model can be developed to **predict house prices based on location and size**.

---

5. Data Cleaning

Data cleaning ensures that the dataset is **accurate, complete, and consistent**.

This stage includes:

- Removing duplicate records
- Handling missing values
- Correcting inconsistent data

Example:
Replacing missing salary values with the **mean salary**.

6. Feature Engineering

Feature engineering is the process of **creating new useful features or selecting important variables** from the dataset to improve model performance.

Activities include:

- Feature selection
- Feature transformation
- Feature extraction

Example:
Combining **date and time into a single feature like "purchase hour"**.

---

7. Data Mining

Data mining is the process of **discovering hidden patterns, relationships, and insights from large datasets** using algorithms and statistical techniques.

Example:

- Customer segmentation
- Market basket analysis
- Fraud detection

The insights obtained help organizations **make better business decisions**.

Q4. What is Data Science? Compare Data Science and Information Science. (6 Marks)

Introduction to Data Science

Data Science is an interdisciplinary field that uses **scientific methods, statistical techniques, algorithms, and computing systems to extract meaningful insights and knowledge from structured and unstructured data.**

It combines concepts from several fields such as:

- Statistics
- Mathematics
- Machine Learning
- Data Analysis
- Computer Science

The main goal of data science is to **analyze large amounts of data and convert it into useful information that can help organizations make better decisions.**

Example:
Companies like **Amazon or Netflix** use data science to analyze user behavior and recommend products or movies.

Data science involves several processes such as:

- Data collection
- Data cleaning
- Data analysis
- Data visualization
- Predictive modeling

These processes help transform **raw data into actionable insights.**

Information Science

Information Science is the study of **collecting, organizing, storing, retrieving, and managing information efficiently**.

It focuses mainly on how information is **structured, stored, processed, and accessed** in information systems.

Information science deals with:

- Information storage
- Information retrieval
- Knowledge organization
- Database systems
- Information management

Example:
Library management systems and search engines use principles of **information science** to organize and retrieve information quickly.

Difference Between Data Science and Information Science

Basis	Data Science	Information Science
Definition	Data science is the field that extracts knowledge and insights from large datasets using statistical and computational methods.	Information science focuses on the collection, organization, storage, and retrieval of information.
Main Objective	To analyze data and discover patterns, predictions, and insights.	To manage and organize information efficiently for easy retrieval.
Focus Area	Data analysis, machine learning, predictive modeling.	Information storage, retrieval systems, and knowledge organization.
Type of Work	Works mainly with large datasets (Big Data).	Works with information systems and databases.
Tools Used	Python, R, machine learning algorithms, data visualization tools.	Database management systems, information retrieval systems.
Output	Produces insights, predictions, and analytical models.	Produces organized and accessible information systems.
Example	Predicting customer buying behavior using machine learning.	Library catalog system that stores and retrieves books information.

Q5. Explain Data Reduction Techniques with Example. (6 Marks)

Introduction to Data Reduction

Data Reduction is a data preprocessing technique used to **reduce the volume of data while maintaining the important information contained in the dataset**. Large datasets require significant storage space and computational power for processing. Therefore, data reduction techniques are used to make data **smaller, simpler, and more efficient to analyze**.

The main objective of data reduction is to **obtain the same analytical results using a reduced representation of the data.**

For example, a company may have a dataset containing **millions of transaction records.** Instead of analyzing the entire dataset, data reduction techniques can summarize the data while preserving useful information.

Data reduction improves:

- Data analysis speed
- Storage efficiency
- Machine learning model performance

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Techniques of Data Reduction

Several techniques are used for reducing data size. The most commonly used techniques are **dimensionality reduction, numerosity reduction, compression, and aggregation**.

---

1. Dimensionality Reduction

Dimensionality reduction is the process of **reducing the number of attributes or features in a dataset while retaining the most important information**.

In many datasets, there may be a large number of features, but not all of them contribute significantly to the analysis. Removing unnecessary or redundant features helps simplify the dataset.

Example:
A dataset containing **100 features** describing customer behavior may be reduced to **15 important features** using dimensionality reduction techniques.

Common methods include:

- Principal Component Analysis (PCA)
- Feature selection techniques

Dimensionality reduction improves **model efficiency and reduces computational complexity**.

---

2. Numerosity Reduction

Numerosity reduction reduces the **number of data records instead of reducing attributes**. In this technique, the original dataset is replaced by a **smaller representation or model** that approximates the original data.

Methods used include:

- Sampling
- Regression models
- Clustering

Example:
Instead of analyzing one million sales records, a **sample of 10,000 records** can be used for analysis.

This method reduces storage requirements and processing time while maintaining similar analytical results.

---

3. Data Compression

Data compression reduces the **size of data by encoding it in a more efficient format**.

Compression techniques can be classified into two types:

- **Lossless compression** – original data can be perfectly reconstructed
- **Lossy compression** – some information is lost during compression

Example:
Text files, images, and videos are often compressed to reduce storage size.

Compression helps improve **storage efficiency and data transmission speed**.

4. Data Aggregation

Data aggregation is the process of **combining multiple data values into a summarized form.**

Instead of storing detailed data, aggregated values are stored.

Example:

Daily sales data:

Day	Sales
Monday	1000
Tuesday	1200
Wednesday	1100

Instead of storing each day's data separately, the weekly total can be stored as:

Total weekly sales = 3300

Aggregation helps simplify data analysis and reduces dataset size.

Advantages of Data Reduction

1. Reduces storage space required for data
2. Improves data processing speed
3. Simplifies data analysis
4. Reduces computational cost for machine learning models
5. Helps manage large datasets efficiently

Q6. What are the Steps of Data Preprocessing? Which Steps May Change the Dimensions of Data? (6 Marks)

Introduction to Data Preprocessing

Data preprocessing is an important step in the data science and data mining process. It involves **transforming raw data into a clean, structured, and usable format for analysis or machine learning algorithms**.

In real-world situations, raw data is often **incomplete, inconsistent, noisy, and unstructured**. If such data is used directly for analysis, it may produce inaccurate results. Therefore, data preprocessing techniques are applied to improve the **quality and reliability of the dataset**.

The main goal of data preprocessing is to **prepare high-quality data that can be efficiently analyzed and used for building predictive models**.

Steps of Data Preprocessing

The data preprocessing process generally includes the following steps:

1. Data Cleaning

Data cleaning is the process of **removing errors, inconsistencies, and noise from the dataset**.

This step improves the quality of data by:

- Handling missing values
- Removing duplicate records
- Correcting incorrect data
- Detecting and removing outliers

Example:
If some entries in the **age column are missing**, they can be replaced with the **mean or median age**.

---

2. Data Integration

Data integration involves **combining data from multiple sources into a single dataset**.

Organizations often collect data from different systems such as:

- Databases
- Spreadsheets
- Web services
- Sensors

Example:
Customer data from a **website database** may be integrated with **sales data from a retail system** to create a complete dataset.

3. Data Transformation

Data transformation converts data into a **suitable format for analysis and machine learning algorithms**.

Common transformation methods include:

- Normalization
- Scaling
- Encoding categorical variables
- Aggregation

Example:
Converting categorical values like **Male/Female** into numerical values such as **0 and 1**.

---

4. Data Reduction

Data reduction is the process of **reducing the size of the dataset while maintaining important information**.

Large datasets require more storage and processing time, so reducing unnecessary data improves efficiency.

Techniques include:

- Dimensionality reduction
- Data compression
- Aggregation
- Sampling

Example:
A dataset with **100 attributes may be reduced to 20 important attributes.**

5. Data Discretization

Data discretization converts **continuous numerical values into discrete categories or intervals.**

Example:

Age values:

18, 25, 32, 40, 55

Can be grouped into:

- 0–20 → Young
- 21–40 → Adult
- 41–60 → Middle Age

Discretization simplifies data and improves certain machine learning algorithms.

Steps That May Change the Dimensions of Data

Some preprocessing steps can **change the dimensions of the dataset** (number of attributes or records).

The main steps that change dimensions are:

1. Data Reduction

Data reduction decreases the number of **features or records** in the dataset.

Example:
Reducing **100 features to 20 features** using dimensionality reduction.

2. Data Transformation

Data transformation can **create new features or modify existing features**, which may change the dataset dimensions.

Example:
Combining two columns like **date and time** into a new feature called **timestamp**.

3. Data Discretization

Discretization changes the **representation of data** by converting continuous attributes into categorical intervals.

Example:
Converting age values into categories such as **young, adult, senior**.

Conclusion

Data preprocessing is an essential step in data analysis and machine learning that prepares raw data for further processing. It includes techniques such as **data cleaning, data integration, data transformation, data reduction, and data discretization**. Some of these steps, particularly **data reduction, transformation, and discretization**, may change the dimensions of the dataset.

Q7. Compare Business Intelligence and Data Science. (6 Marks)

Introduction

Organizations today use data to make better decisions and improve their operations. Two important fields that help analyze and utilize data are **Business Intelligence (BI)** and **Data Science**.

Both fields work with data, but they differ in their **objectives, techniques, and applications**. Business Intelligence mainly focuses on **analyzing historical data to support business decision-making**, while Data Science focuses on **analyzing large datasets and building predictive models to forecast future outcomes**.

Business Intelligence

Business Intelligence (BI) refers to the technologies, tools, and processes used to **collect, analyze, and present historical business data** to help organizations make informed decisions.

BI systems mainly focus on:

- Data reporting
- Data visualization
- Dashboards
- Business performance monitoring

Example:
A company may use BI tools to analyze **monthly sales reports and customer transactions** to understand past performance.

Common BI tools include:

- Power BI
- Tableau
- SAP Business Objects

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Data Science

Data Science is an interdisciplinary field that uses **statistical methods, machine learning algorithms, and computational techniques to extract insights and predictions from large datasets**.

Unlike BI, data science focuses on **predictive and advanced analytics**.

Data science involves processes such as:

- Data collection
- Data cleaning
- Data analysis
- Machine learning modeling
- Predictive analytics

Example:
An online shopping platform may use data science to **predict which products a customer is likely to buy next**.

Common tools used in data science include:

- Python
- R
- Machine learning algorithms
- Big data frameworks

Difference Between Business Intelligence and Data Science

Basis	Business Intelligence	Data Science
Definition	Business Intelligence focuses on analyzing historical data to support business decision making.	Data Science focuses on extracting insights and building predictive models using large datasets.
Type of Analysis	Descriptive analysis (what happened in the past).	Predictive and prescriptive analysis (what will happen in the future).
Data Type	Mostly structured data from databases.	Works with structured, semi-structured, and unstructured data.
Techniques Used	Reporting, dashboards, data visualization.	Machine learning, statistical analysis, predictive modeling.
Goal	Improve business decision-making using past data.	Discover patterns and predict future outcomes.
Tools Used	Power BI, Tableau, SQL, Excel.	Python, R, TensorFlow, machine learning tools.
Example	Analyzing monthly sales performance using dashboards.	Predicting future product demand using machine learning.

Q8. Explain the 5 V's of Big Data with Examples. (6 Marks)

Introduction to Big Data

Big Data refers to **extremely large and complex datasets that cannot be easily processed using traditional data processing tools**. With the growth of digital technologies such as social media, sensors, and online transactions, the amount of data generated every day has increased tremendously.

Big Data is characterized by five important properties known as the **5 V's of Big Data**. These are:

- Volume
- Velocity
- Variety
- Veracity
- Value

These characteristics describe the **nature, challenges, and importance of Big Data**.

1. Volume

Volume refers to the **huge amount of data generated and stored by organizations every day**.

Modern systems generate enormous volumes of data from various sources such as:

- Social media platforms
- Online transactions
- Sensors and IoT devices
- Mobile applications

Example:
Companies like **Facebook, Google, and Amazon** generate terabytes or even petabytes of data every day.

Handling such large amounts of data requires **special storage systems and distributed computing technologies** like Hadoop and cloud storage.

---

2. Velocity

Velocity refers to the **speed at which data is generated, collected, and processed.**

In many modern applications, data is produced continuously and must be processed in **real time or near real time.**

Example:

- Stock market transactions
- Online payment systems
- Social media updates

For instance, **Twitter generates thousands of tweets every second**, which must be processed quickly to analyze trends.

High data velocity requires technologies capable of **real-time data processing and streaming analytics.**

3. Variety

Variety refers to the **different types and formats of data generated from multiple sources.**

Unlike traditional systems that mainly deal with structured data, Big Data includes:

- Structured data (databases, tables)
- Semi-structured data (JSON, XML)
- Unstructured data (images, videos, text, audio)

Example:

An online shopping platform collects various types of data such as:

- Customer purchase records (structured data)
- Product reviews (text data)
- Images of products (multimedia data)

Managing and analyzing such diverse data formats is one of the major challenges of Big Data.

---

4. Veracity

Veracity refers to the **accuracy, reliability, and quality of data**.

Big Data often contains **incomplete, inconsistent, or noisy information**, which can affect the accuracy of analysis.

Example:

Social media data may contain:

- Fake information
- Duplicate records
- Incorrect user inputs

Therefore, data scientists must perform **data cleaning and preprocessing** to improve data quality before analysis.

---

5. Value

Value refers to the **useful insights and benefits that organizations obtain from analyzing Big Data**.

Collecting large volumes of data is useful only if it can generate **meaningful insights that support decision-making**.

Example:

- Retail companies analyze customer purchase data to improve marketing strategies
- Healthcare organizations analyze medical data to improve patient care
- Banks analyze transaction data to detect fraud

The ultimate goal of Big Data is to **extract v ble information that can help businesses grow and innovate**.

Q9. Explain Data Types and Levels of Measurement with Suitable Examples. (6 Marks)

Introduction

In data science and statistics, data can be classified based on its **structure and measurement levels**. Understanding different types of data helps data scientists choose appropriate **analysis techniques, statistical methods, and machine learning algorithms**.

Data is generally categorized into two main classifications:

1. **Types of Data based on structure**
2. **Levels of Measurement**

---

1. Types of Data Based on Structure

Data can be classified into **structured, semi-structured, and unstructured data** depending on how it is organized and stored.

1. Structured Data

Structured data refers to **data that is organized in a fixed format such as rows and columns**. It is easy to store, process, and analyze using traditional database systems.

Structured data is usually stored in **relational databases or spreadsheets**.

Examples include:

- Customer records in a database
- Sales transactions in tables
- Employee information stored in spreadsheets

Example table:

ID	Name	Age	Salary
101	Rahul	30	50000
102	Priya	28	45000

This format makes structured data easy to query using **SQL and database systems**.

2. Semi-Structured Data

Semi-structured data does not follow a strict tabular format but still contains **some organizational structure such as tags or key-value pairs**.

It is more flexible than structured data.

Common examples include:

- XML files
- JSON files
- Email messages

Example (JSON format):

```
{
 "Name": "Rahul",
 "Age": 30,
 "City": "Pune"
}
```

Semi-structured data is widely used in **web applications and data exchange systems**.

---

3. Unstructured Data

Unstructured data refers to **data that does not have a predefined format or structure**.

It is difficult to store and analyze using traditional database systems.

Examples include:

- Images
- Videos
- Audio files
- Social media posts
- Text documents

Example:
Photos uploaded on Instagram or videos uploaded on YouTube are examples of **unstructured data**.

Analyzing such data usually requires **advanced technologies such as machine learning and natural language processing**.

2. Levels of Measurement

Levels of measurement describe **how data values are categorized and interpreted** in statistical analysis. The four levels of measurement are **nominal, ordinal, interval, and ratio**.

---

1. Nominal Scale

Nominal data represents **categories or labels without any specific order**.

The values are used only for classification and cannot be ranked.

Examples include:

- Gender (Male, Female)
- Blood group (A, B, AB, O)
- Country names

In nominal data, we can perform operations like **counting frequencies**.

---

2. Ordinal Scale

Ordinal data represents **categories that have a meaningful order or ranking**, but the difference between values is not fixed.

Examples include:

- Education level (Primary, Secondary, Higher)
- Customer satisfaction (Low, Medium, High)
- Movie ratings (1 star, 2 stars, 3 stars)

Here, we know the **order of categories**, but cannot measure the exact difference between them.

3. Interval Scale

Interval data represents **numerical data where the difference between values is meaningful, but there is no true zero point**.

Examples include:

- Temperature measured in Celsius or Fahrenheit
- Calendar dates

Example:
The difference between **20°C and 30°C is the same as between 30°C and 40°C**, but 0°C does not mean absence of temperature.

---

4. Ratio Scale

Ratio data is the **most informative level of measurement**. It has all the properties of interval data and also includes a **true zero point**, which represents the absence of the quantity.

Examples include:

- Height
- Weight
- Age
- Income

Example:
If one person earns **₹40,000 and another earns ₹20,000**, we can say the first person earns **twice as much**.

Q10. What Does Feature Engineering Typically Include? (6 Marks)

Introduction to Feature Engineering

Feature Engineering is the process of **selecting, transforming, and creating relevant features (variables) from raw data to improve the performance of machine learning models**.

In data science, raw data collected from different sources may not be directly suitable for building predictive models. Therefore, feature engineering is performed to **extract meaningful information from the data and convert it into a form that machine learning algorithms can understand effectively**.

Good feature engineering can significantly **improve model accuracy, reduce training time, and enhance predictive performance**.

Feature engineering mainly includes the following activities:

- Feature Selection
- Feature Extraction
- Feature Encoding

1. Feature Selection

Feature selection is the process of **selecting the most relevant and important features from the dataset while removing unnecessary or redundant features**.

Large datasets often contain many attributes, but not all of them contribute significantly to the prediction process. Removing irrelevant features helps reduce **model complexity and computational cost**.

Example:

A dataset predicting house prices may contain many attributes such as:

- House color
- Number of rooms
- Location
- Area size
- Distance from city center

Among these, features like **number of rooms, location, and area size** may be more important than house color.

Therefore, feature selection helps choose **only the most useful variables** for the model.

Advantages of feature selection:

- Reduces overfitting
- Improves model performance
- Reduces training time

2. Feature Extraction

Feature extraction is the process of **creating new features from existing data by transforming or combining the original variables.**

Instead of simply selecting features, this method **generates new informative features** that represent the data more effectively.

Example:

Suppose a dataset contains the following attributes:

- Date of purchase
- Time of purchase

These two features can be combined to create a new feature called **purchase hour**, which may help analyze customer behavior.

Another example:

In image processing, important features such as **edges, shapes, or patterns** can be extracted from images.

Feature extraction helps reduce data dimensionality while preserving useful information.

---

3. Feature Encoding

Feature encoding is the process of **converting categorical data into numerical format so that machine learning algorithms can process it**.

Many machine learning algorithms require numerical input, but datasets often contain categorical values such as:

- Gender (Male, Female)
- Product category (Electronics, Clothing, Grocery)

These categorical variables must be converted into numbers.

Common encoding techniques include:

Label Encoding

Each category is assigned a numerical value.

Example:

Category	Encoded Value
Male	0
Female	1

One-Hot Encoding

Each category is converted into separate binary columns.

Example:

Product	Electronics	Clothing
Electronics	1	0
Clothing	0	1

Encoding helps machine learning algorithms interpret categorical information correctly.

Importance of Feature Engineering

Feature engineering is essential because:

1. It improves the accuracy of machine learning models.
2. It helps remove irrelevant or redundant features.
3. It reduces the complexity of datasets.
4. It enables algorithms to understand categorical data.
5. It helps discover hidden patterns in the data.

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Q11. Describe Different Ways of Handling Missing Data with Suitable Example. (6 Marks)

Introduction

In real-world datasets, it is very common to encounter **missing values**. Missing data occurs when some values in a dataset are **not recorded, unknown, or unavailable**. Missing data can negatively affect data analysis and machine learning models because many algorithms require **complete datasets** to function properly.

Therefore, handling missing data is an important step in **data preprocessing and data cleaning**. Proper handling of missing values improves the **accuracy and reliability of analysis results**.

There are several techniques used to handle missing data.

Methods for Handling Missing Data

1. Ignoring or Deleting Missing Values

One of the simplest methods is to **ignore or remove records that contain missing values**. This method is also called **listwise deletion**.

In this approach, rows with missing data are removed from the dataset.

Example:

Name	Age	Salary
Rahul	30	50000
Priya	—	45000
Amit	28	42000

The row containing the missing age value for **Priya** may be removed from the dataset.

Advantages:

- Easy to implement
- No need for complex calculations

Disadvantage:

- Useful data may be lost if many records contain missing values.

2. Filling Missing Values with Mean

In this method, missing values are replaced with the **mean (average) value of that attribute**.

Example:

If the age values in a dataset are:

25, 30, 35, —, 40

Mean age = (25 + 30 + 35 + 40) / 4 = **32.5**

The missing value can be replaced with **32.5**.

This method works best for **numerical data without extreme outliers**.

---

3. Filling Missing Values with Median

The **median** is the middle value when the data is arranged in order. Replacing missing values with the median is useful when the dataset contains **outliers or skewed data**.

Example:

Dataset:

10, 12, 15, 18, 100

The mean is heavily influenced by the value **100**, but the median is **15**, which better represents the dataset.

Therefore, missing values can be replaced with the **median value**.

4. Filling Missing Values with Mode

The **mode** is the most frequently occurring value in the dataset. This method is commonly used for **categorical data**.

Example:

Dataset of colors:

Red, Blue, Blue, Green, Blue

Mode = **Blue**

If a color value is missing, it can be replaced with **Blue**.

---

5. Predicting Missing Values

In this method, missing values are estimated using **machine learning or statistical models**.

Techniques such as **regression analysis, decision trees, or clustering** can be used to predict missing values based on other attributes in the dataset.

Example:

If the salary value is missing for an employee, it may be predicted based on:

- Age
- Education
- Work experience

This method is more accurate but requires **advanced analysis techniques**.

Example of Handling Missing Data

Consider a dataset containing employee information:

Name	Age	Salary
Rahul	30	50000
Priya	28	—
Amit	35	60000

The missing salary value for **Priya** could be handled by:

- Removing the row
- Replacing it with the **mean salary**
- Predicting the salary using regression

Conclusion

Handling missing data is a critical step in data preprocessing because missing values can reduce the accuracy of data analysis and machine learning models. Techniques such as **ignoring missing values, replacing them with mean, median, or mode, and predicting missing values using models** help ensure that the dataset remains reliable and suitable for analysis.

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Applications of Data Science in Different Domains

1. Healthcare

In healthcare, data science is used to analyze medical records, patient data, and diagnostic results to improve treatment and healthcare services.

Applications include:

- Disease prediction
- Medical image analysis
- Personalized treatment recommendations
- Drug discovery

Example:
Hospitals use data science to **predict the risk of diseases such as diabetes or heart disease** based on patient data.

---

2. Finance

Financial institutions use data science to analyze large volumes of transaction data and identify patterns.

Applications include:

- Fraud detection
- Credit risk analysis
- Stock market prediction
- Customer behavior analysis

Example:
Banks use machine learning algorithms to **detect fraudulent transactions in real time.**

3. E-commerce

Online shopping platforms use data science to understand customer preferences and improve user experience.

Applications include:

- Product recommendation systems
- Customer segmentation
- Demand forecasting
- Pricing optimization

Example:
E-commerce companies like Amazon recommend products based on **customer purchase history and browsing behavior**.

Data Science in Urban Planning

Urban planning involves the **design and management of cities and infrastructure to improve the quality of life for citizens**. Data science helps city planners analyze data related to transportation, population, energy usage, and environmental conditions.

Using data science, governments can develop **smart city solutions** that improve urban living conditions.

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Problems in Urban Planning that Can Be Solved Using Data Science

1. Traffic Management

Data science can analyze traffic data collected from sensors, cameras, and GPS devices to **reduce traffic congestion and optimize traffic signals.**

Example:
Cities can use real-time traffic data to adjust signal timings and reduce travel time.

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2. Public Transportation Optimization

By analyzing passenger data and travel patterns, authorities can improve **bus and metro routes and schedules.**

Example:
Increasing the number of buses on routes where passenger demand is high.

---

3. Waste Management

Data science helps analyze waste generation patterns and optimize **waste collection routes and schedules.**

Example:
Smart bins equipped with sensors can notify authorities when they are full.

---

4. Energy Management

Cities can use data science to monitor and analyze **energy consumption patterns** and improve energy efficiency.

Example:
Smart grids can optimize electricity distribution based on demand.

---

5. Urban Development Planning

Data science helps analyze population growth and housing data to plan **better infrastructure and housing projects**.

Example:
Identifying areas that require new schools, hospitals, or roads.

Conclusion

Data science has many applications across different industries, helping organizations make **data-driven decisions and solve complex problems**. In urban planning, data science plays a crucial role in improving **traffic management, public transportation, waste management, energy usage, and infrastructure planning**, ultimately contributing to the development of smarter and more sustainable cities.

Q13. Differentiate between Data Science, Artificial Intelligence, and Machine Learning. (6 Marks)

Introduction

Modern technologies such as **Data Science, Artificial Intelligence (AI), and Machine Learning (ML)** are closely related fields that deal with analyzing data and building intelligent systems. Although they are interconnected, they have **different goals, methods, and applications**.

Data Science focuses on **extracting insights from data**, Artificial Intelligence focuses on **creating intelligent systems that can perform tasks similar to humans**, and Machine Learning is a **subset of AI that allows systems to learn automatically from data**.

Understanding the differences between these three fields is important for selecting the right techniques for solving real-world problems.

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Difference Between Data Science, Artificial Intelligence, and Machine Learning

Basis	Data Science	Artificial Intelligence	Machine Learning
Definition	Data Science focuses on extracting insights and knowledge from large datasets.	Artificial Intelligence focuses on creating intelligent systems that mimic human intelligence.	Machine Learning is a subset of AI that enables systems to learn automatically from data.
Main Objective	Analyze data and generate useful insights for decision-making.	Build systems capable of intelligent behavior.	Develop algorithms that learn patterns from data.
Scope	Broad field combining statistics, data analysis, and machine learning.	Broad field that includes machine learning, robotics, and expert systems.	Subset of AI focusing on learning from data.
Techniques Used	Data analysis, statistical modeling, data visualization.	Search algorithms, reasoning systems, neural networks.	Regression, classification, clustering algorithms.
Dependency	Uses machine learning and statistical techniques.	May include machine learning and other AI techniques.	Part of Artificial Intelligence.
Example	Predicting customer buying behavior using data analysis.	Self-driving cars and intelligent assistants.	Spam email detection using classification algorithms.